

Technology News

Solar power without solar cells

A new discovery about the magnetic effect of light could lead to solar power without traditional semiconductor-based solar cells.

Researchers at the University of Michigan have discovered that at the right intensity of 10 million watts per square centimetre, light, when travelling through a material such as glass that does not conduct electricity, can generate magnetic fields that are a hundred million times stronger than previously thought possible. The resulting magnetic force could replace the electric effect exploited by the current technology, paving the way for 'optical batteries.'

"This could lead to a new kind of solar cell without semiconductors and without absorption to produce charge separation. In solar cells, the light goes into a material, gets absorbed and creates heat. Here we expect to have a very low-heat load. Instead of the light being absorbed, energy is stored in the magnetic moment. Intense magnetisation can be induced by intense light and then it is ultimately capable of providing a capacitive power source," said Stephen Rand, professor in the departments of electrical engineering and computer science, physics and applied physics.

Mind-controlled games

Iceland-based MindGames has developed games that you play with your state of mind, not with your fingers. You need to wear a headset—a simplified version of the electroencephalogram (EEG) which is commonly used in hospitals—to measure the brain activity and play. One of the interesting things about using brainwave headsets to control games is that you can theoretically use 'relaxation' to control a character in an exciting situ-



W.I.L.D. game app from MindGames for iPhone and iPad

ation. For example, you could relax in order to make your car go faster in a racing game.

One game app, dubbed W.I.L.D., is rolled out for iPhone and iPad. By concentrating and relaxing, players navigate dream landscapes and complete tasks like putting out fires, bending spoons and creating galaxies.

Another app, 'Tug of Mind,' purports to train users how to relax under pressure. After uploading a headset

of, say, an ex-boyfriend or mother-in-law, the photo is transformed into an angry *avatar* that screams insults (you prerecord the message). The *avatar* starts smiling if the player remains calm. Keep it up and the game is won.

Spintronic computer chips closer to reality

Researchers have developed the first electronic circuit that merges traditional inorganic semiconductors with